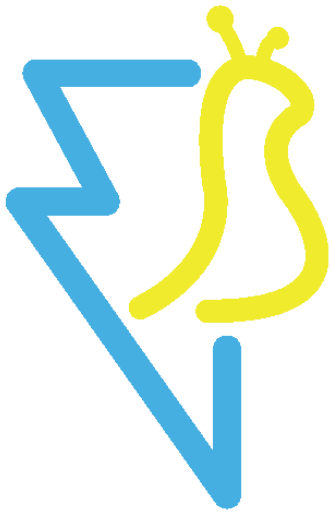


Formula Slug

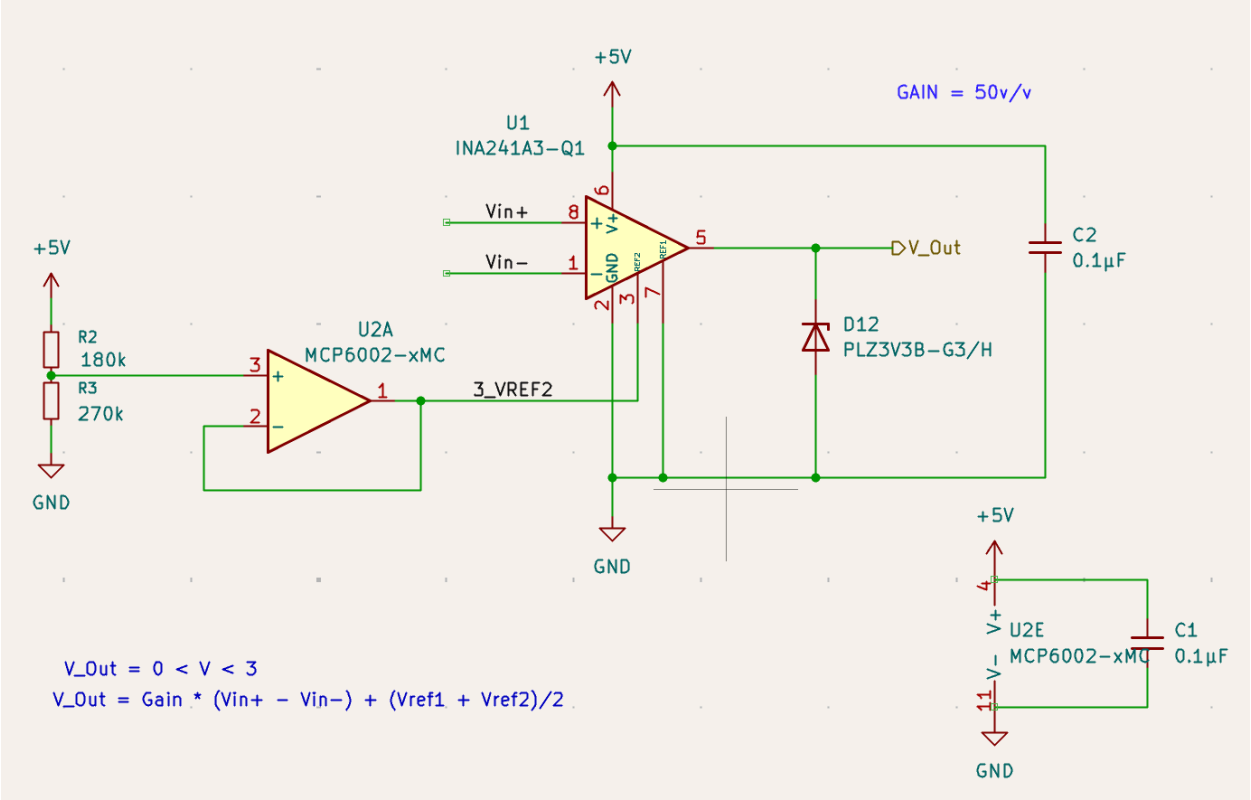
PE Onboarding: Bidirectional Shunt

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By Jonathan Palioca

Dear Edmund,

I'm going to level with you brother. I have spent the past two weeks racking my brain trying to come up with good solutions for a shunt but every way has had a trade off. This solution, while nowhere near perfect, is the one I spent the most time and effort researching and drawing out.



This is the INA241A3-Q1 bidirectional current sense ic with a gain of 50v/v which would be reading the voltage difference of a 50µΩ shunt resistor with a reference voltage of 3v. As configured I had planned to place it on the high side which comes with a few benefits (it can measure a short in the system, and does not obstruct the ground plane). The busbar shunt will have a power output max of 18w, well within the 50w, and with the size it shouldn't cause any problems.

Running in this configuration it measures from -600A to 600A on a respective 0 < V < 3 scale where zero amps results in a Vout = 1.5v. The utility of this configuration comes with the ref2 pin. By shorting the op amps non inverting terminal to +5v, the ref2 pin will be receiving around

5v which shifts the output voltage scale. This new scale (0 < V < 5), outputs a zero amp reading of 2.5v. On the previous scale this 2.5v equates to a current of 400 amps!!!. With the BSPD set to 5kW and the relays calibrated to the original scales voltage, this 2.5v reading should be well over the threshold required to trip the relays. I should also say, I have connected a zener diode to the output to avoid the unlikely event of more than 3.3v coming out of V_out. This would protect the MCU especially while the reference plane is shifted, although I don't think the battery will have much current flowing during testing.

$$V_{out} = Gain * (I_{shunt} * R_{Shunt}) + (V_{ref1} + V_{ref2})/2$$

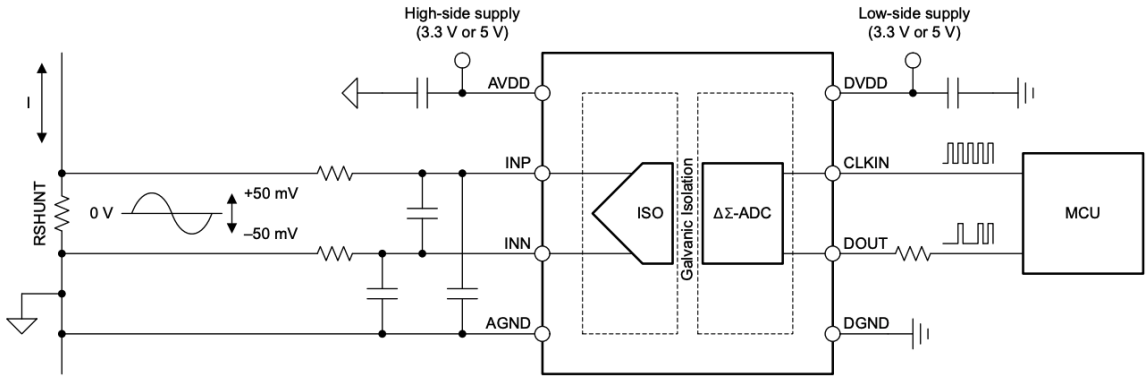
$$V_{ref1} = 0$$

$$I_{shunt} = \frac{V_{out} - (V_{ref2}/2)}{Gain(R_{shunt})}$$

Where this setup runs into a wall is its lack of transient protection. Being run on the high side of the system it is subjected to the full voltage of the 108v battery. With a common voltage max of 110v it gives very little headroom from the little INA241 chip. I spent time looking into TVS diodes as a solution but with the battery voltage max being so close to the chip's max voltage I couldn't find any that would make much of a difference. Even with the IC's max rating of 120v it is at the mercy of the transients. Unless somehow the system didn't spike like that 👀. But I'm doubtful…

FLOAT THE FRONT END!!

If Transient spikes are a big issue for a highside current shunt, there are two main ways to get around it. The safest way to protect the sensitive circuitry from these spikes would be to use an IC such as the AMC0306M05 which galvanically isolates the sensitive components from the current sensing hardware.



Typical Application

This however comes at the cost of… well cost, less accuracy, and to comply with BSPD testing, the hall effects sensor would need to be used in tandem to simulate the current.

The other option is to place the INA241 circuit on the low side of the system. Since the common mode voltage it will be experiencing wont be the full 108v the amplifier will have a lot more headroom to operate safely. This however comes with its own downsides, it loses its ability to detect short circuits, and it can shift/disrupt the ground plane. This wouldn't be too much of an issue, unless the shunt were to fail open which is highly improbable. On the bright side, I believe this could be run without the hall effects sensor as you would still be able to short the op amp to supply the ref2 pin 5v.

All in all I'd like to say that I had a great time working on this project, and I really appreciate the time you took to help us out. I hope you have a good weekend 🙏